COMPUTER AIDED ENGINEERING GRAPHICS

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Experiment No. 6

Projection of Lines - II

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Group Name

I B.Tech I Semester

Computer Science Engineering

VARDHAMAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Affiliated to **JNTUH,** Approved by **AICTE,** Accredited by **NAAC** with **A++** Grade, **ISO 9001:2015** Certified

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Projection of Lines II

1 Objectives

1. Understand the concepts of true length, apparent length, and true inclination of a straight line with respect to the horizontal plane (H.P.) and vertical plane (V.P.).
2. Apply AutoCAD drawing commands to construct the projections of lines inclined to one reference plane and parallel to the other, and of lines situated in H.P. or V.P.
3. Analyze the relationship between true length, apparent lengths in the front and top views, and the angle of inclination to determine unknown parameters of a line.

2 Pre-Lab Questions

Q1. Explain the difference between Scale and Stretch Commands in AUTOCAD?

Q2. List and explain the different types of the Array command available in AutoCAD.

Q3. If a Line is inclined to H.P. and parallel to V.P. State the appearance of its top view and front view.

Q4. If a Line is inclined to V.P. and parallel to H.P. State the appearance of its top view and front view.

3 Aim

Perform the following in AUTOCAD

- A 60 *mm* long line AB has end A 20 *mm* above H.P. and 40 *mm* in front of the V.P. The line is inclined at 30^0 to the H.P. and is parallel to the V.P. Draw the projections of the line.
- An 70 *mm* long line LM is inclined at 30^0 to the V.P. and is parallel to the H.P. The end L of the line is 15 *mm* above the H.P. and 30 *mm* in front of the V.P. Draw the projections of the line.

4 Software Used

- AUTOCAD

5 System Requirements

1. **System Type:** Windows 10 64 Bit Operating System
2. **Processor:** i3
3. **RAM:** 4 GB

6 Commands Used

7 Experimental procedure

8 Learning Outcomes

At the end of this experiment, the student will be able to:

1. Recall the terms true length, apparent length, inclination with H.P., and inclination with V.P.
2. Explain how the inclination of a line to one reference plane affects the length and orientation of its front view and top view when the line is parallel to the other plane.
3. Construct accurate AutoCAD projections for lines inclined to H.P. (parallel to V.P.), lines inclined to V.P. (parallel to H.P.), and lines lying in H.P. or V.P.
4. Determine the true length and true inclination of a line from the given apparent lengths and angles in its projections.
5. Evaluate the accuracy of the drawn projections by comparing the apparent lengths with the true length and verifying the inclination angles.

9 Post-Lab Questions

Q1. The line AB of 100 mm length is inclined at an angle of 30° to the H.P. and parallel to the V.P. The point A is 15 mm above H.P. and 20 mm in front of V.P. Draw the front view and top view of the line.

Q2. A 60 mm long line AB is parallel to and 20 mm in front of the V.P. The ends A and B are 10 mm and 50 mm above the H.P. respectively. Draw the projections of the line and determine its inclination with the H.P. Also, locate the traces.

Q3. The length of the top view of a line parallel to the V.P. and inclined at 45° to the H.P. is 50 mm. One end of the line is 12 mm above the H.P. and 25 mm in front of the V.P. Draw the projections of the line and determine its true length.

Q4. The front view of a line inclined at 30° to the V.P. is 65 mm long. Draw the projections and true length of the line when it is parallel to and 40 mm above the H.P. and its one end is 30 mm in front of the V.P.

Q5. A 60 mm long line is inclined at 45° to the V.P. The vertical trace (V.T.) is 25 mm above XY and the horizontal trace (H.T.) does not exist. Draw the projections of the line when an end is 10 mm in front of the V.P.

Q6. An electric switch and a bulb fixed on a wall are 5 m apart. The distance between them measured parallel to the floor is 4 metres. If the switch is 1.5 m above the floor, find the height of the bulb and the inclination of the line joining the two with the floor.

10 Result

The solutions of the projection of lines are analysed and drawn in the AUTOCAD.